

Understanding Linux Containers

LXC or Linux containers are OS level virtualisations that run several isolated applications on the parent control host system. Through containerisation, developers can package applications with their dependencies so that the apps can run on other host systems flawlessly.



Linux containers are applications that are isolated from the host system on which they are running. Developers are allowed by the containers to package an application through the libraries and dependencies needed for the application, and ship it as one package. For developers and sysadmins, the transfer of code from the development environments into production is rapid and replicable.

Containers are similar to virtual machines. With containers, one does not have to replicate a whole operating system. Containers only need individual components in order to operate. This boosts performance and reduces the size of the application.

A perfect example to understand containers

To understand containers better, just imagine that you are developing an application. You do your work on a laptop and your environment has a specific configuration, which might be different from that of other developers. The application you're developing relies on that configuration and is dependent on specific files. Meanwhile, your business has test and production environments which are standardised with their own configurations and their own sets of supporting files.

You want to emulate those environments as much as possible locally but without all of the overhead of recreating the server environments. So, how do you make your app work across these environments, pass quality assurance and get your app deployed without massive headaches, rewriting or break-fixing? The answer is: containers.

The container that holds your application has the necessary configuration (and files) so that you can move it from development, to testing and on to production seamlessly. So a crisis is averted and everyone is happy!

That's a basic example, but there are many different ways in which Linux containers can be applied to problems, when solutions for ultimate portability, configurability and isolation are needed. It doesn't matter whether the infrastructure is on-premise, on the cloud or a hybrid of the two -- containers are the solution. Of course, choosing the right container platform is just as important as the containers themselves.

How do I orchestrate containers?

Orchestrators manage the nodes and containers manage the work load with the goal of maintaining a stable and scalable environment. This includes auto-scaling and self-